

Towards prevention of vitamin D deficiency and beyond: knowledge gaps and research needs in vitamin D nutrition and public health.

The North American Institute of Medicine (IOM) recently published their report on dietary reference intakes (DRI) for Ca and vitamin D. The DRI committee's deliberations underpinning this most comprehensive report on vitamin D nutrition to date benefited hugely from a much expanded knowledge base in vitamin D over the last decade or more. However, since their release, the vitamin D DRI have been the subject of intense controversy, which is largely due to the persistence of fundamental knowledge gaps in vitamin D. These can be identified at the levels of exposure, metabolism, storage, status, dose-response, function and beneficial or adverse health effects, as well as safe and effective application of intake recommendations at the population level through sustainable food-based approaches. The present review provides a brief overview of the approach used by the IOM committee to revise the DRI for vitamin D and to collate from a number of authoritative sources key knowledge gaps in vitamin D nutrition from the public health perspective. A number of research topics are outlined and data requirements within these are identified and mapped to the risk assessment framework used by the DRI committee. While not intended as an exhaustive list, it provides a basis for organising and prioritising research efforts in the area of vitamin D, which may offer a perspective on the major areas in need of attention. It is intended to be of use to researchers, national policy makers, the public health community, industry groups and other relevant stakeholders including funding institutions.